(MATH000) Mathematical Language

Notation and Conventions

Jungwoo Kim
Department of Mathematics, Pohang University of Science and Technology

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Part I. General Mathematical Language

1 General Conventions

These conventions apply throughout. Widely used standard alternatives may be listed for the same concept in preferred order; the first is used throughout.

Item	Convention
Scalars	Ordinary mathematical italic letters, for example a, b, c, λ, t.
Vectors	Bold lowercase letters, for example $\mathbf{u}, \mathbf{v}, \mathbf{x}$. Unless otherwise stated, vectors are column vectors.
Matrices	Bold uppercase letters, for example $\mathbf{A}, \mathbf{B}, \mathbf{M}$.
Dimension subscripts	Prefer dimension subscripts when the dimension is readily available and useful; omit them when it is clear, immaterial, or not readily available rather than inferring a dimension solely to supply a subscript.
Greek vectors and matrices	Bold Greek letters when a Greek symbol denotes a vector or matrix, for example $\boldsymbol{\xi}$ and $\boldsymbol{\Phi}$.
Number systems	Blackboard bold, for example $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$.
Sets	Ordinary uppercase letters such as A, B, X, Y, unless a field-specific convention requires otherwise.
Families and collections	Calligraphic letters when useful, for example $\mathcal{F}, \mathcal{T}, \mathcal{B}$.
Named operators	Upright Roman type, for example det, ker, im, rk, and tr.
Large operators	Display size in displayed mathematics and notation tables; sums and products place indices above and below, limits place subscripts below, and integral bounds retain standard right-side placement.
Greek variants	Distinct glyphs represent distinct symbols; in particular, ε and ϵ are not interchangeable.

Notation	Meaning
$(a_1, \ldots, a_n)$	Ordered tuple; parentheses are also the default matrix delimiters.
(a, b)	Open interval.
$[a, b]$	Closed interval.
$[a, b)$	Left-closed, right-open interval.
$(a, b]$	Left-open, right-closed interval.
$\{a_1, \ldots, a_n\}$	Set delimiters.
$\{x \in A \mid P(x)\}$	Set-builder notation.
$\langle u, v \rangle$	Pairing or inner product, according to context.
$\lvert x \rvert$	Absolute value, modulus, cardinality, or order, according to context.
$\lVert x \rVert$	Norm.
s.t.	Abbreviation for “such that” when naturally required.
$a = b$	Equality.
$a \coloneqq b$	a is defined to be b.
$a \equiv b$	Identity, equivalence, or congruence, according to context.
$a \cong b$	Isomorphism or congruence, according to context.
$a \approx b$	Approximation.
$f \colon X \to Y$	Function or map from X to Y.
$x \mapsto f(x)$	Elementwise rule of a map.
$g \circ f$	Composition: first f, then g.
$f\vert_A,\ f\restriction_A$	Restriction of f to A.

Notation	Meaning
id_X	Identity map on X.

2 Theorem-like Environments

Item	Convention
Definition	Precise meaning of a term, object, property, or construction.
Theorem	Principal mathematical result.
Proposition	Useful result of more limited scope.
Lemma	Supporting result, usually for a later proof.
Corollary	Result following readily from a preceding result.
Conjecture	Statement proposed to be true but not proved in full generality.
Hypothesis	Named assumption or proposed statement.
Formula	Distinguished computational identity or closed form.
Algorithm	Explicit step-by-step procedure.
Fact	Verified statement not requiring theorem status.
Example	Illustration of a definition, construction, phenomenon, or result.
Counterexample	Example disproving a general assertion or showing a hypothesis is necessary.
Remark	Supplementary context, warning, or qualification.
Problem	Mathematical task intended to be solved.
Open Problem	Problem not known to be solved in the stated form.
Idea	Decisive conceptual insight.
Strategy	Broad problem-solving plan.
Tactic	Local step within a larger strategy.

Proof labels are written as *Proof* and solution labels as *Solution*, each followed by a period.

Notation	Meaning
$\square$	Ending for ordinary proofs and solutions.
$\blacksquare$	Ending for substantial, difficult, or particularly important proofs.
Q.E.D.	Ending for exceptionally difficult, significant, or paper-level proofs.

Enumeration and Subdivision. Enumeration labels are set in upright Roman type. Their form is determined by structural role rather than nesting depth.

Item	Convention
1.1, 1.2, 1.3, ...	Independent child problems or standalone units under a common parent; each may be referenced separately.
(1), (2), (3), ...	Ordinary enumerated items, such as choices, cases, examples, or procedural steps.
(a), (b), (c), ...	Connected subparts of a single problem sharing a common setup.
(i), (ii), (iii), ...	Formal mathematical clauses, such as hypotheses, conditions, properties, or equivalent statements within definitions, theorems, propositions, and related statements.

The distinction between 1.1 and (a) is independence: 1.1 denotes a standalone child unit, whereas (a) denotes a connected subpart of one problem. Likewise, (1) is used for ordinary enumeration, while (i) is reserved for formal mathematical clauses. Nested enumeration preserves these roles; for example, a subpart (a) may contain formal clauses (i), (ii), ...

Item	Convention
Displayed mathematics	Remains part of the surrounding sentence and receives punctuation when grammar requires it.
Equation numbering	Number equations that are referenced later or benefit from a stable label.
Logical prose	Use words such as "and" and "or" unless symbolic logic improves clarity.
Abbreviations	Use established abbreviations only when standard and unambiguous.

Part II. Notation by Mathematical Area

3 Set Theory and Logic

Uppercase letters usually denote sets; lowercase letters usually denote their elements.

Notation	Meaning
$\mathbb{N} = \{1, 2, 3, \dots\}$	Positive integers.
$\mathbb{N}_0 = \{0, 1, 2, 3, \dots\}$	Nonnegative integers.
$\mathbb{Z}$	Integers.
$\mathbb{Q}$	Rational numbers.
$\mathbb{R}$	Real numbers.
$\mathbb{C}$	Complex numbers.
$\varnothing,\ \emptyset$	Empty set.
$x \in A$	x is an element of A.
$x \notin A$	x is not an element of A.
$A \subseteq B$	A is a subset of B; equality is allowed.
$A \subsetneq B$	A is a proper subset of B.
$A \supseteq B$	A is a superset of B; equality is allowed.
$A \supsetneq B$	A is a proper superset of B.
$A \cup B$	Union.
$A \cap B$	Intersection.
$\displaystyle\bigcup_{i \in I} A_i$	Union of an indexed family of sets.
$\displaystyle\bigcap_{i \in I} A_i$	Intersection of an indexed family of sets.
$A \setminus B,\ A - B$	Set difference.
$A \triangle B,\ A \mathbin{\Delta} B$	Symmetric difference.
A^c	Complement of A in the ambient set.
$\mathcal{P}(A),\ 2^A$	Power set of A.
$A \times B$	Cartesian product.
$A \sqcup B,\ A \mathbin{\dot\cup} B$	Disjoint union.
$A \cap B = \varnothing$	A and B are disjoint.
$\lvert A \rvert,\ \#A$	Cardinality.
$\{x \in A \mid P(x)\}$	Elements $x \in A$ satisfying $P(x)$.
$R \subseteq X \times Y$	Binary relation between X and Y.
xRy	x is related to y by the relation R.

Notation	Meaning
R^{-1}	Inverse relation of R.
$[x]_R,\ [x],\ R[x]$	Equivalence class of x under R; abbreviated forms are used when the relation is clear.
$x \sim y$	Equivalent elements under an understood equivalence relation.
$A/{\sim}$	Set of equivalence classes.
$\forall x \in A$	For every $x \in A$.
$\exists x \in A$	There exists $x \in A$.
$\exists! x \in A$	There exists a unique $x \in A$.
$\nexists x \in A$	There does not exist $x \in A$.
$P \Rightarrow Q$	P implies Q.
$P \Leftarrow Q$	Q implies P.
$P \nRightarrow Q$	P does not imply Q.
$P \Leftrightarrow Q$	P if and only if Q.
$P \wedge Q$	Logical conjunction.
$P \vee Q$	Logical disjunction.
$\neg P$	Logical negation.
$\operatorname{dom} f$	Domain of f.
$\operatorname{codom} f$	Codomain of f.
$f(A)$	Image of A under f.
$f^{-1}(B)$	Preimage of B under f.
$\operatorname{im} f$	Image of the map f.
$f\colon X \hookrightarrow Y$	Injective map.
$f\colon X \twoheadrightarrow Y$	Surjective map.
$f\colon X \hookrightarrow\!\!\!\!\rightarrow Y$	Bijective map.
$f\colon X \xrightarrow{\sim} Y$	Isomorphism; for sets, equivalently a bijection.
$\mathbb{1}_A(x)$	Indicator function of A.
$\aleph_0$	Cardinality of a countably infinite set.
$\mathfrak{c}$	Cardinality of the continuum.

4 Calculus and Analysis

The differential d is upright; ε is the default small positive quantity in analysis. Function spaces take their domains in parentheses, for example $C([a,b])$, $\mathcal{R}([a,b])$, and $L^p(\Omega)$.

Notation	Meaning
$\lim\limits_{x\to a} f(x)$	Limit as x approaches a.
$\lim\limits_{x\to a^-} f(x)$	Left-hand limit as x approaches a.
$\lim\limits_{x\to a^+} f(x)$	Right-hand limit as x approaches a.
$\min A$	Minimum of A.
$\max A$	Maximum of A.
$\inf A$	Infimum of A.
$\sup A$	Supremum of A.
$f'(a)$	First derivative of f at a.
$f^{(n)}(a)$	nth derivative of f at a.
$\dfrac{\mathrm{d}y}{\mathrm{d}x}$	Derivative of y with respect to x.

Notation	Meaning
$\dfrac{\partial f}{\partial x_i},\ \partial_i f$	Partial derivative with respect to x_i.
$\displaystyle\int_a^b f(x)\ \mathrm{d}x$	Definite integral over $[a, b]$.
$\displaystyle\iint_D f\ \mathrm{d}A$	Double integral over D.
$\displaystyle\iiint_\Omega f\ \mathrm{d}V$	Triple integral over Ω.
$\displaystyle\oint_C f(z)\ \mathrm{d}z$	Contour integral over a closed curve C.
$\displaystyle\sum_{k=1}^{n} a_k$	Finite sum.
$\displaystyle\sum_{k=1}^{\infty} a_k$	Infinite series.
$\displaystyle\prod_{k=1}^{n} a_k$	Finite product.
$e^x,\ \exp x$	Exponential function.
$\ln x,\ \log_e x$	Logarithm to base e.
$\lg x,\ \log_{10} x$	Logarithm to base 10.
$\mathrm{lb}\, x,\ \log_2 x$	Logarithm to base 2.
$\log_a x$	Logarithm to base a.
$\sin x$	Sine.
$\cos x$	Cosine.
$\tan x$	Tangent.
$\csc x$	Cosecant.
$\sec x$	Secant.
$\cot x$	Cotangent.
$\arcsin x,\ \sin^{-1} x$	Principal inverse sine.
$\arccos x,\ \cos^{-1} x$	Principal inverse cosine.
$\arctan x,\ \tan^{-1} x$	Principal inverse tangent.
$\operatorname{arccsc} x,\ \csc^{-1} x$	Principal inverse cosecant.
$\operatorname{arcsec} x,\ \sec^{-1} x$	Principal inverse secant.
$\operatorname{arccot} x,\ \cot^{-1} x$	Principal inverse cotangent.
$\sinh x$	Hyperbolic sine.
$\cosh x$	Hyperbolic cosine.
$\tanh x$	Hyperbolic tangent.
$\operatorname{csch} x$	Hyperbolic cosecant.
$\operatorname{sech} x$	Hyperbolic secant.
$\coth x$	Hyperbolic cotangent.

Notation	Meaning
$\operatorname{arcsinh} x,\ \operatorname{arsinh} x,\ \sinh^{-1} x$	Principal inverse hyperbolic sine.
$\operatorname{arccosh} x,\ \operatorname{arcosh} x,\ \cosh^{-1} x$	Principal inverse hyperbolic cosine.
$\operatorname{arctanh} x,\ \operatorname{artanh} x,\ \tanh^{-1} x$	Principal inverse hyperbolic tangent.
$\operatorname{arccsch} x,\ \operatorname{arcsch} x,\ \operatorname{csch}^{-1} x$	Principal inverse hyperbolic cosecant.
$\operatorname{arcsech} x,\ \operatorname{arsech} x,\ \operatorname{sech}^{-1} x$	Principal inverse hyperbolic secant.
$\operatorname{arccoth} x,\ \operatorname{arcoth} x,\ \coth^{-1} x$	Principal inverse hyperbolic cotangent.
$\Gamma(z)$	Gamma function.
$B(x, y)$	Beta function.
$\zeta(s)$	Riemann zeta function.
$D\mathbf{F}(\mathbf{a})$	Total derivative; Jacobian matrix in coordinates.
$\operatorname{Jac}(\mathbf{F})$	Jacobian determinant.
$\operatorname{Hess} f$	Hessian matrix of f.
∇f	Gradient of f.
$\nabla \cdot \mathbf{F}$	Divergence of $\mathbf{F}$.
$\nabla \times \mathbf{F}$	Curl of $\mathbf{F}$.
Δf	Laplacian of f.
$\mathbf{r}(t)$	Parametrized curve.
$\mathrm{d}s$	Element of arc length.
$\int_C \mathbf{F} \cdot \mathrm{d}\mathbf{r}$	Line integral along C.
$\iint_S \mathbf{F} \cdot \mathbf{n}\ \mathrm{d}S$	Flux integral through S.
$x_n \to x$	Convergence of a sequence.
$a_n \nearrow L$	Monotone increasing convergence to L.
$a_n \searrow L$	Monotone decreasing convergence to L.
$\limsup a_n$	Limit superior.
$\liminf a_n$	Limit inferior.
$f_n \to f,\ f_n \xrightarrow{\text{ptw}} f$	Pointwise convergence.
$f_n \rightrightarrows f,\ f_n \xrightarrow{\text{unif}} f$	Uniform convergence.
$f_n \xrightarrow{\text{a.e.}} f$	Convergence almost everywhere.
$C(\Omega),\ C^0(\Omega)$	Continuous functions on Ω.
$C^k(\Omega)$	k times continuously differentiable functions on Ω.
$C^\infty(\Omega)$	Smooth functions on Ω.
$\mathcal{R}([a, b])$	Riemann-integrable functions on $[a, b]$.
$L^p(\Omega)$	Lebesgue L^p space on Ω.
$L^\infty(\Omega)$	Lebesgue space of essentially bounded functions on Ω.
$\\|f\\|_\infty$	Supremum norm $\sup_{x\in\Omega} \lvert f(x)\rvert$, when finite.
$\\|f\\|_{L^p(\Omega)},\ \\|f\\|_p$	L^p-norm $\left(\int_\Omega \lvert f(x)\rvert^p\ \mathrm{d}x\right)^{1/p}$ for $1 \le p < \infty$.
$\\|f\\|_{L^\infty(\Omega)}$	L^∞-norm $\operatorname{ess\,sup}_{x\in\Omega} \lvert f(x)\rvert$.

Notation	Meaning
$B_r(x)$	Open ball of radius r centered at x.
$\overline{B}_r(x)$	Closed ball of radius r centered at x.
$z = x + iy$	Complex number with real part x and imaginary part y.
$\operatorname{Re} z$	Real part of z.
$\operatorname{Im} z$	Imaginary part of z; distinct from lowercase im for image.
$\overline{z}$	Complex conjugate.
$\lvert z\rvert$	Complex modulus.
$\arg z$	Argument of z.
$\operatorname{Res}(f; a),\ \operatorname{Res}_{z=a} f(z)$	Residue of f at a.

5 Linear Algebra

Notation	Meaning
$\mathbf{A} = (a_{ij})$	Matrix with (i, j)-entry a_{ij}.
$\mathbf{A}^\mathsf{T}$	Transpose of $\mathbf{A}$.
$\mathbf{A}^\mathsf{H}$	Conjugate transpose of $\mathbf{A}$.
$\mathbf{A}^{-1}$	Inverse of $\mathbf{A}$.
$\mathbf{I}_n,\ \mathbf{I}$	Identity matrix; the dimension subscript is preferred when readily available and useful.
$\mathbf{0}$	Zero vector or zero matrix, with dimension understood.
$\mathbf{D}$	Diagonal matrix when a local name is required.
$\mathbf{P}$	Projection matrix when a local name is required.
$\operatorname{diag}(d_1, \ldots, d_n)$	Diagonal matrix with entries $d_1, \ldots, d_n$.
$\mathbf{1}$	All-ones vector.
$e^{\mathbf{A}},\ \exp(\mathbf{A})$	Matrix exponential; $e^{t\mathbf{A}}$ is the standard time-dependent form.
$\det(\mathbf{A}),\ \lvert\mathbf{A}\rvert$	Determinant.
$\operatorname{rk}(\mathbf{A}),\ \operatorname{rank}(\mathbf{A})$	Rank.
$\operatorname{tr}(\mathbf{A})$	Trace.
$\operatorname{adj}(\mathbf{A})$	Adjugate matrix.
$\operatorname{Col}(\mathbf{A})$	Column space.
$\operatorname{Row}(\mathbf{A})$	Row space.
$\operatorname{Null}(\mathbf{A}),\ \ker \mathbf{A}$	Null space.
$\ker T$	Kernel of a linear map T.
$\operatorname{im} T$	Image of a linear map T.
$\mathbf{e}_i$	ith standard basis vector.
$\dim V$	Dimension of V.
$\operatorname{Span}(\mathbf{v}_1, \ldots, \mathbf{v}_k)$	Span of the listed vectors.
$U \oplus V$	Direct sum.
$\langle \mathbf{u}, \mathbf{v}\rangle,\ (\mathbf{u}, \mathbf{v})$	Inner product; over complex spaces, conjugate-linear in the first argument and linear in the second.
$\mathbf{u} \cdot \mathbf{v},\ \mathbf{u}^\mathsf{T}\mathbf{v}$	Standard Euclidean inner product on $\mathbb{R}^n$.
$\mathbf{u}^\mathsf{H}\mathbf{v}$	Standard Hermitian inner product on $\mathbb{C}^n$.
$\lVert\mathbf{v}\rVert$	Norm of $\mathbf{v}$.
$\mathbf{u} \perp \mathbf{v}$	Orthogonality.
$U^\perp$	Orthogonal complement of the subspace U.

Notation	Meaning
$\operatorname{proj}_U \mathbf{x}$	Orthogonal projection of $\mathbf{x}$ onto the subspace U.
$\operatorname{proj}_{\mathbf{v}} \mathbf{x}$	Orthogonal projection of $\mathbf{x}$ onto $\operatorname{Span}(\mathbf{v})$.
$\mathbf{u} \times \mathbf{v}$	Cross product in $\mathbb{R}^3$.
$p_{\mathbf{A}}(\lambda),\ \chi_{\mathbf{A}}(\lambda)$	Characteristic polynomial of $\mathbf{A}$.
$m_{\mathbf{A}}(\lambda),\ \mu_{\mathbf{A}}(\lambda)$	Minimal polynomial of $\mathbf{A}$.
$\lambda,\ \lambda_i$	Eigenvalue of $\mathbf{A}$.
$\mathbf{v},\ \mathbf{v}_i$	Eigenvector of $\mathbf{A}$.
$E_\lambda(\mathbf{A}),\ V_\lambda(\mathbf{A})$	Eigenspace of $\mathbf{A}$ associated with λ.
$\sigma(\mathbf{A}),\ \operatorname{spec}(\mathbf{A})$	Spectrum of $\mathbf{A}$.
$\rho(\mathbf{A})$	Spectral radius.
$\sigma_i(\mathbf{A})$	ith singular value of $\mathbf{A}$.

6 Differential Equations

The Fourier transform uses the normalization

$$\widehat{f}(\boldsymbol{\xi}) \coloneqq \int_{\mathbb{R}^n} f(\mathbf{x}) e^{-2\pi i \mathbf{x}\cdot\boldsymbol{\xi}}\, \mathrm{d}\mathbf{x}, \qquad f(\mathbf{x}) = \int_{\mathbb{R}^n} \widehat{f}(\boldsymbol{\xi}) e^{2\pi i \mathbf{x}\cdot\boldsymbol{\xi}}\, \mathrm{d}\boldsymbol{\xi},$$

whenever the Fourier inversion formula applies.

Notation	Meaning
y'	First derivative with respect to the understood independent variable.
y''	Second derivative.
$\dot{x}$	First derivative of x with respect to time.
$\ddot{x}$	Second derivative of x with respect to time.
$y_h,\ y_c$	Homogeneous or complementary part of a solution.
y_p	Particular solution.
$c_1, c_2, \ldots$	Arbitrary constants in a general solution.
$W(y_1, \ldots, y_n)(t)$, $W[y_1, \ldots, y_n](t)$	Wronskian of $y_1, \ldots, y_n$.
$\mathbf{x}' = \mathbf{A}\mathbf{x} + \mathbf{g}(t)$, $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{g}(t)$	Linear first-order system in state-vector form.
$\boldsymbol{\Phi}(t)$	Fundamental matrix of a linear system.
$\mathcal{L}\{f(t)\}(s)$, $\mathcal{L}[f](s)$	Laplace transform of f.
$\mathcal{L}^{-1}\{F(s)\}(t)$, $\mathcal{L}^{-1}[F](t)$	Inverse Laplace transform of F.
$(f * g)(t)$	One-sided convolution, $(f * g)(t) \coloneqq \int_0^t f(\tau) g(t-\tau)\, \mathrm{d}\tau$.
$H(t-a),\ u(t-a)$	Heaviside or unit-step function shifted to $t = a$; its value at the jump is specified when relevant.
$G(x, \xi)$	Green's function; the variables depend on context.
$J_\nu(x)$	Bessel function of the first kind of order ν.
$Y_\nu(x),\ N_\nu(x)$	Bessel function of the second kind (Neumann function) of order ν.
$\widehat{f}(\boldsymbol{\xi}),\ \mathcal{F}f(\boldsymbol{\xi})$	Fourier transform under the normalization above.

7 Combinatorics and Discrete Mathematics

Notation	Meaning
$\binom{n}{r},\ {}_nC_r$	Number of r-element subsets of an n-element set.
$\left(\!\binom{n}{r}\!\right),\ {}_nH_r$	r-combinations with repetition from n types; equals $\binom{n+r-1}{r}$.
$n!$	Factorial.
$n^{\underline{k}},\ (n)_k$	Falling factorial.
$n^{\overline{k}},\ n^{(k)}$	Rising factorial.
$O(g(n))$	Big-O upper asymptotic bound.
$o(g(n))$	Little-o strict asymptotic bound.
$\Theta(g(n))$	Two-sided asymptotic order.
$\Omega(g(n))$	Asymptotic lower bound.
$(X, \preceq)$	Partially ordered set (poset).
$\prec,\ \succeq,\ \succ$	Strict and reverse order relations associated with $\preceq$.
$\leq_{\text{lex}}$	Lexicographic order.
$x \lessdot y$	Cover relation: y covers x.
B_n	Boolean lattice (Boolean poset) on an n-element ground set.
$G = (V, E)$	Graph with vertex set V and edge set E.
$V(G)$	Vertex set of G.
$E(G)$	Edge set of G.
$N(v),\ \Gamma(v)$	Neighborhood of vertex v.
$\deg_G(v),\ \deg(v)$	Degree of vertex v; the graph subscript may be omitted when clear.
$\deg^+(v),\ \deg^-(v)$	Out-degree and in-degree of v in a directed graph.
$d_G(u, v),\ d(u, v)$	Graph distance between vertices u and v.
K_n	Complete graph on n vertices.
$K_{m,n}$	Complete bipartite graph with parts of sizes m and n.
P_n	Path graph on n vertices.
C_n	Cycle graph on n vertices.
$\mathbf{A}(G),\ \mathbf{A}_G$	Adjacency matrix of G.
$\mathbf{L}(G),\ \mathbf{L}_G$	Graph Laplacian, usually $\mathbf{D}(G) - \mathbf{A}(G)$.
$G - e$	Graph obtained by deleting the edge e.
$G/e,\ G : e$	Graph obtained by contracting the edge e; G/e is the preferred notation.
$T(G),\ \tau(G)$	Number of spanning trees of G; $T(G)$ is preferred here to avoid conflict with the standard number-theoretic $\tau(n)$.
$r(k, \ell),\ r(k)$	Ramsey numbers; $r(k) = r(k, k)$.
$A(x) = \sum_{n=0}^{\infty} a_n x^n$	Ordinary generating function of the sequence $(a_n)_{n\geq 0}$.

8 Probability and Statistics

Probability and expectation are written in blackboard bold by default.

Notation	Meaning
$\mathbb{P}(A),\ P(A)$	Probability of event A.
$\mathbb{P}(A \mid B),\ P(A \mid B)$	Conditional probability of A given B.

Notation	Meaning
$X \perp\!\!\!\perp Y$	X and Y are independent.
$X \perp\!\!\!\perp Y \mid Z$	X and Y are conditionally independent given Z.
$\mathbb{E}[X],\ E[X]$	Expectation of random variable X.
$\mathbb{E}[X \mid Y],\ E[X \mid Y]$	Conditional expectation of X given Y.
$\mathrm{Var}(X)$	Variance of X.
$\mathrm{Cov}(X, Y)$	Covariance of X and Y.
$X \sim \mathrm{Bin}(n, p)$	Binomial distribution.
$X \sim \mathrm{Pois}(\lambda)$	Poisson distribution.
$X \sim \mathcal{N}(\mu, \sigma^2)$, $X \sim N(\mu, \sigma^2)$	Normal distribution with mean μ and variance σ^2.
$X_n \xrightarrow{\text{a.s.}} X$	Almost-sure convergence.
$X_n \xrightarrow{\mathbb{P}} X$	Convergence in probability.
$X_n \xrightarrow{L^p} X$	Convergence in L^p.
$X_n \xrightarrow{\text{d}} X,\ X_n \Rightarrow X$	Convergence in distribution.

9 Geometry

In triangle notation, A, B, C denote vertices and a, b, c the opposite side lengths.

Notation	Meaning
$\triangle ABC$	Triangle with vertices A, B, C.
$\angle ABC$	Angle with vertex B.
$\overline{AB}$	Segment joining A and B.
AB	Length of $\overline{AB}$.
$[ABC]$	Area of $\triangle ABC$.
$\mathrm{length}(C),\ \ell(C)$	Length of C.
$\mathrm{area}(S)$	Area of S.
$\mathrm{vol}(S)$	Volume of S.
$\mathrm{dist}(P, S)$	Distance from P to S.
$\mathrm{conv}(S)$	Convex hull of S.
$\mathbb{R}^n$	n-dimensional real Euclidean space.
$\mathbb{H}^n$	n-dimensional hyperbolic space.
$\parallel$	Parallel.
$\perp$	Perpendicular or orthogonal.

10 Number Theory

Notation	Meaning
$a \mid b$	a divides b.
$a \nmid b$	a does not divide b.
$\gcd(a, b)$	Greatest common divisor, taken nonnegative.
$\mathrm{lcm}(a, b)$	Least common multiple.
$a \equiv b \pmod{n}$	a is congruent to b modulo n.
$[a]_n$	Residue class of a modulo n.
$\mathbb{Z}/n\mathbb{Z}$	Ring of residue classes modulo n.

Notation	Meaning
$(\mathbb{Z}/n\mathbb{Z})^\times$	Multiplicative group of units modulo n.
$a^{-1} \pmod n$	Multiplicative inverse of a modulo n.
$\left(\dfrac{a}{p}\right)$	Legendre symbol.
$\varphi(n)$	Euler totient function.
$\mathrm{ord}_n(a)$	Multiplicative order of a modulo n.
$v_p(n),\ \nu_p(n)$	p-adic valuation of n.
$\tau(n)$	Number of positive divisors of n.
$\sigma(n)$	Sum of positive divisors of n.
$\mu(n)$	Möbius function.
$\mathrm{rad}(n)$	Product of the distinct prime divisors of n.
$\pi(x)$	Number of primes not exceeding x.
$\mathbb{F}_q,\ \mathrm{GF}(q)$	Finite field with q elements.

11 Algebra

G, H usually denote groups, R, S rings, and F, K, L fields.

Notation	Meaning
G	Group.
$e,\ 1_G$	Identity element of a group.
$\lvert G\rvert$	Order of a finite group.
$\langle S\rangle$	Subgroup generated by S.
$H \le G$	H is a subgroup of G.
$H \trianglelefteq G$	H is a normal subgroup of G.
$[G : H]$	Index of H in G.
G/H	Quotient group.
$\ker\varphi$	Kernel of φ.
$\mathrm{im}\,\varphi$	Image of φ.
$\mathrm{Aut}(G)$	Automorphism group of G.
$Z(G)$	Center of G.
$C_G(x)$	Centralizer of x in G.
$\mathrm{GL}_n(F)$	General linear group over F.
$\mathrm{SL}_n(F)$	Special linear group over F.
$S_n,\ \mathfrak{S}_n$	Symmetric group on n symbols.
$A_n,\ \mathfrak{A}_n$	Alternating group on n symbols.
R	Ring.
$I \trianglelefteq R$	I is an ideal of R.
$(a_1, \ldots, a_n),\ \langle a_1, \ldots, a_n\rangle$	Ideal generated by $a_1, \ldots, a_n$.
R/I	Quotient ring.
F	Base field in an algebraic context.
K/F	Field extension of F by K.
$F[x]$	Polynomial ring in x over F.
$\deg f$	Degree of a polynomial f.
$\mathrm{char}\, F$	Characteristic of F.
$\mathbb{H}$	Quaternion division algebra over $\mathbb{R}$.
$\mathrm{Gal}(L/K)$	Galois group of L/K.

Notation	Meaning
$\operatorname{trdeg}_K L$	Transcendence degree of L over K.

12 Topology

Notation	Meaning
$(X, \mathcal{T})$	Topological space X with topology $\mathcal{T}$.
A°, $\operatorname{int} A$	Interior of A.
$\overline{A}$, $\operatorname{cl} A$	Closure of A.
∂A	Boundary of A.
$\pi_1(X, x_0)$	Fundamental group of X based at x_0.
$\chi(X)$	Euler characteristic of X.
$X \cong Y$	Homeomorphic spaces.
$X \simeq Y$	Homotopy equivalent spaces.

13 Miscellaneous Notation

Notation	Meaning
$\lfloor x \rfloor$	Floor of x.
$\lceil x \rceil$	Ceiling of x.
$\{x\}$	Fractional part $x - \lfloor x \rfloor$, when unambiguous.
$\operatorname{sgn}(x)$	Sign of a real number or permutation, according to context.
Δq	Finite change in a quantity q.
ϵ_{ijk}	Levi–Civita symbol.